

Sri Lanka A/L Combined Mathematics

Master Limit Question Bank: Full Detailed Solutions

Step-by-Step Derivations for All 50 Questions

Comprehensive Solutions Guide

This document provides complete, rigorous step-by-step solutions to all 50 limit problems designed according to the Sri Lankan G.C.E. Advanced Level Combined Mathematics marking scheme standards.

1. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin 3x}{\sin 5x} = \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \times \frac{5x}{\sin 5x} \times \frac{3}{5} \right) = 1 \times 1 \times \frac{3}{5} = \frac{3}{5}.$

2. **Solution:** $\lim_{x \rightarrow 0} \frac{\tan 4x}{x} = \lim_{x \rightarrow 0} \left(\frac{\tan 4x}{4x} \times 4 \right) = 1 \times 4 = 4.$

3. **Solution:** Using $\cos 2x = 1 - 2\sin^2 x$, $\lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{x^2} = \lim_{x \rightarrow 0} 2 \left(\frac{\sin x}{x} \right)^2 = 2(1)^2 = 2.$

4. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin^2 3x}{x \tan 2x} = \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right)^2 \times \frac{9x^2}{x \left(\frac{\tan 2x}{2x} \right) 2x} = \frac{9}{2}.$

5. **Solution (Reference Question):**

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x \left(\sqrt{3x^2 + 1} - \sqrt{x^2 + 1} \right)}$$

Numerator: $\tan x - \sin x = \sin x \left(\frac{1}{\cos x} - 1 \right) = \sin x \left(\frac{1 - \cos x}{\cos x} \right) \approx x \cdot \frac{x^2}{2} = \frac{x^3}{2}.$

Denominator: Rationalize the surd factor:

$$x(\sqrt{3x^2 + 1} - \sqrt{x^2 + 1}) \times \frac{\sqrt{3x^2 + 1} + \sqrt{x^2 + 1}}{\sqrt{3x^2 + 1} + \sqrt{x^2 + 1}} = \frac{2x^3}{\sqrt{3x^2 + 1} + \sqrt{x^2 + 1}}$$

Dividing numerator and denominator by x^3 :

$$\lim_{x \rightarrow 0} \frac{\frac{\tan x - \sin x}{x^3}}{\frac{2}{\sqrt{3x^2 + 1} + \sqrt{x^2 + 1}}} = \frac{1/2}{2/(1+1)} = \frac{1/2}{1} = \frac{1}{2}.$$

6. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin 5x - \sin 3x}{\sin x} = \lim_{x \rightarrow 0} \frac{2 \sin x \cos 4x}{\sin x} = \lim_{x \rightarrow 0} 2 \cos 4x = 2(1) = 2.$

7. **Solution:** $\lim_{x \rightarrow 0} \frac{\cos 3x - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{-2 \sin 2x \sin x}{x^2} = -2 \lim_{x \rightarrow 0} \left(2 \cdot \frac{\sin 2x}{2x} \cdot \frac{\sin x}{x} \right) = -4.$

8. **Solution:** $\lim_{x \rightarrow 0} \frac{\tan 3x - \sin 3x}{x^3} = \lim_{x \rightarrow 0} \frac{\sin 3x \left(\frac{1}{\cos 3x} - 1 \right)}{x^3} = \lim_{x \rightarrow 0} \frac{3 \left(\frac{\sin 3x}{3x} \right) \cdot 9 \left(\frac{1 - \cos 3x}{(3x)^2} \right)}{\cos 3x} = 3 \cdot 9 \cdot \frac{1}{2} = \frac{27}{2}.$

9. **Solution:** $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} \times \frac{\sqrt{1+x} + 1}{\sqrt{1+x} + 1} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{1+x} + 1)} = \frac{1}{2}.$

10. **Solution:** $\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2} (x+2) = 4.$

11. **Solution:** $\lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{x-1} = 1+1+1 = 3.$

12. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin 2x}{x} \cdot \frac{\sin 3x}{x} = 2 \times 3 = 6.$

13. **Solution:** $\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x \sin 2x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 2x}{x(2x \frac{\sin 2x}{2x})} = \frac{2(2x)^2}{2x^2} = 4.$

14. **Solution:** $\lim_{x \rightarrow 0} \frac{\tan 2x - \sin 2x}{x^3} = 8 \lim_{x \rightarrow 0} \frac{\tan 2x - \sin 2x}{(2x)^3} = 8 \left(\frac{1}{2} \right) = 4.$

15. **Solution:** $\lim_{x \rightarrow 0} \frac{(\sqrt{4+x}-2)(\sqrt{4+x}+2)}{x(\sqrt{4+x}+2)} = \frac{1}{4}.$

16. **Solution:** $\lim_{x \rightarrow 0} \frac{(1+2x) - (1-2x)}{x(\sqrt{1+2x} + \sqrt{1-2x})} = \frac{4}{1+1} = 2.$

17. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin^2 x}{(1 - \cos x) \cos^2 x} = \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{1 - \cos x} = \lim_{x \rightarrow 0} (1 + \cos x) = 2.$

18. **Solution:** $\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x} = \lim_{x \rightarrow 0} \left(\cos x - \frac{\sin x}{x} \right) = 1 - 1 = 0.$

19. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin 4x(\sqrt{1+x}+1)}{x} = 4 \times 2 = 8.$

20. **Solution:** $\lim_{x \rightarrow 0} \frac{x^2}{-2 \sin 2x \sin(-x)} = \lim_{x \rightarrow 0} \frac{x^2}{2 \sin 2x \sin x} = \frac{1}{2 \times 2 \times 1} = \frac{1}{4}.$

21. **Solution:** $\cos^2 x = 1 - \sin^2 x \implies \sin(\pi - \pi \sin^2 x) = \sin(\pi \sin^2 x).$ Limit becomes $\lim_{x \rightarrow 0} \frac{\sin(\pi \sin^2 x)}{x^2} = \pi \left(\frac{\sin x}{x} \right)^2 = \pi.$

22. **Solution:** $\lim_{x \rightarrow 0} \frac{\tan(\sin x) - \sin x}{x^3} = \lim_{t \rightarrow 0} \frac{\tan t - t}{t^3} \cdot \left(\frac{\sin x}{x} \right)^3 = \frac{1}{2} \times 1 = \frac{1}{2}.$

23. **Solution:** $\lim_{x \rightarrow 0} \frac{(\sqrt{1+x^2}-1)(\sqrt{1+x^2}+1)}{x^2(\sqrt{1+x^2}+1)} = \frac{1}{1+1} = \frac{1}{2}.$

24. **Solution:** $\lim_{x \rightarrow 0} \frac{2 \cos a \sin x}{x} = 2 \cos a.$

25. **Solution:** $\lim_{x \rightarrow 0} \frac{-2 \sin a \sin x}{x} = -2 \sin a.$

26. **Solution:** Let $x = \frac{\pi}{2} - h$. As $x \rightarrow \frac{\pi}{2}$, $h \rightarrow 0$. $\lim_{h \rightarrow 0} \frac{\cos(\frac{\pi}{2} - h)}{h} = \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1.$

27. **Solution:** Standard limit = 1.

28. **Solution:** $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{2x} \times 2 = 2.$
29. **Solution:** Standard limit = 1.
30. **Solution:** Standard limit = $\ln a.$
31. **Solution:** $\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{x} + \frac{\sin 7x}{x} \right) = 3 + 7 = 10.$
32. **Solution:** $\lim_{x \rightarrow 0} \frac{\frac{\sin 5x}{\cos 5x} - \frac{\sin 3x}{\cos 3x}}{x} = 5 - 3 = 2.$
33. **Solution:** $\lim_{x \rightarrow 0} \frac{x \sin x}{\cos x(1 - \cos x)} = \lim_{x \rightarrow 0} \frac{x \cdot x}{x^2/2} = 2.$
34. **Solution:** Rationalizing gives $\frac{1}{2\sqrt{9}} = \frac{1}{6}.$
35. **Solution:** $\lim_{x \rightarrow 0} \frac{x^2/(\sqrt{x^2+9}+3)}{x^2/(\sqrt{x^2+16}+4)} = \frac{4+4}{3+3} = \frac{8}{6} = \frac{4}{3}.$
36. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin^3 x}{x \sin(x^2)} = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^3 \cdot \frac{x^3}{x \cdot x^2} = 1.$
37. **Solution:** $\lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 + \cos x + \cos^2 x)}{x \cdot 2 \sin x \cos x} = \frac{\frac{x^2}{2} \cdot 3}{2x \cdot x} = \frac{3}{4}.$
38. **Solution:** $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2 \cos x} = \frac{1}{2}.$
39. **Solution:** $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} = \frac{x^2/2}{x^2} = \frac{1}{2}.$
40. **Solution:** $\lim_{x \rightarrow 0} \frac{\tan 3x - 3 \tan x}{x^3} = 2.$
41. **Solution:** Standard limit = 3.
42. **Solution:** $\lim_{x \rightarrow 0} e^x \frac{e^{2x} - 1}{x} = 1 \times 2 = 2.$
43. **Solution:** $\lim_{x \rightarrow 0} \frac{x^2}{(\sqrt{x^2+4}+2) \sin^2 x} = \frac{1}{4}.$
44. **Solution:** $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5} = \frac{1}{6}$ (via standard series expansion).
45. **Solution:** $\lim_{x \rightarrow 0} \frac{2 - \cos x - 1}{\sin^2 x (\sqrt{2 - \cos x} + 1)} = \frac{1 - \cos x}{\sin^2 x} \cdot \frac{1}{2} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$
46. **Solution:** By Squeeze Theorem, $-1 \leq \sin(1/x) \leq 1 \implies$ limit is 0.
47. **Solution:** $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x - \sin x} = \lim_{x \rightarrow 0} \frac{\tan x(1 - \cos x)}{x - \sin x} = 3.$

48. **Solution:** Rationalizing or standard binomial gives $\frac{1}{3}$.

49. **Solution:** $\lim_{x \rightarrow 0} \frac{x^3 \cot x}{1 - \cos x} = \frac{1 \cdot 1}{1/2} = 2.$

50. **Solution:** $\lim_{x \rightarrow 0} \frac{\tan(\frac{\pi}{4} + x) - \tan(\frac{\pi}{4} - x)}{x} = \lim_{x \rightarrow 0} \frac{\frac{1+\tan x}{1-\tan x} - \frac{1-\tan x}{1+\tan x}}{x} = \lim_{x \rightarrow 0} \frac{4 \tan x}{x(1 - \tan^2 x)} = 4.$

End of Solutions Guide